Table 6.1 Market Observations from Jan. 2000—Jun. 2012

Statistic	Avg Volatility		Avg Correlation		Avg Dispersion	
	LC	SC	LC	SC	LC	SC
Avg:	36%	54%	32%	23%	36%	58%
Stdev:	16%	20%	15%	12%	16%	20%
Min:	18%	30%	3%	2%	16%	33%
Max:	119%	146%	81%	67%	118%	142%

illustrates the higher volatility period after the flash crash in May 2010 and again during the US debt crisis in fall 2011. Volatility appears to be time varying with a clustering effect. Thus, traditional models that give the same weighting to all historical observations may not be the most accurate representation of actual volatility. The volatility clustering phenomenon was the reason behind the advanced volatility modeling techniques such as ARCH, GARCH, and EWMA. These models are further discussed below. Table 6.1 provides a summary of our market observations over the period from January 2000 through June 2012.

FORECASTING STOCK VOLATILITY

In this section, we describe various volatility forecasting models as well as appropriate techniques to estimate their parameters. We also introduce a new model that incorporates a historical measure coupled with insight from the derivatives market. These models are:

- Historical Moving Average (HMA)
- Exponential Weighted Moving Average (EWMA)
- Autoregressive Models (ARCH and GARCH)
- HMA-VIX Adjustment

Some of these descriptions and our empirical findings have been disseminated in the Journal of Trading's "Intraday Volatility Models: Methods to Improve Real-Time Forecasts," Fall 2012. Below we follow the same outline and terminology as in the journal.

Volatility Models

We describe four different volatility models: the historical moving average (HMA), the exponential weighted moving average (EWMA) introduced by JP Morgan (1996), an autoregressive heteroscedasticity (ARCH) model introduced by Engle (1982), a generalized autoregressive